# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 7 — SCIENCE · HARD LEVEL MCQ TEST SET 2

**Full Syllabus (Chapters 1–11)** | RBSE Pattern | Session 2026–27

**Time: 1 Hour** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

## MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

*Choose the correct option and write it in your answer book.*

**1.** A train leaves at 6 a.m. and runs at a steady 60 km/h. By 12 noon it has covered: 1

**(a)** 720 km **(b)** 120 km **(c)** 60 km **(d)** 360 km

**2.** A person breathes 16 times a minute, taking in 500 mL each time. The volume of air in one minute is: 1

**(a)** 16 mL **(b)** 8000 mL **(c)** 4000 mL **(d)** 500 mL

**3.** Which energy change occurs in respiration? 1

**(a)** heat to chemical **(b)** sound to chemical

**(c)** light to chemical **(d)** chemical energy to usable energy

**4.** During a lunar eclipse, the Moon passes into Earth's: 1

**(a)** shadow **(b)** atmosphere only **(c)** orbit only **(d)** light

**5.** The Moon takes 27.3 days to complete a revolution but 29.5 days from full moon to full moon because: 1

**(a)** the Moon slows down **(b)** the Earth moves around the Sun meanwhile

**(c)** the Sun moves **(d)** the Moon stops

**6.** Which statement about milk of magnesia is correct? 1

**(a)** it is a neutral salt used as a drink **(b)** it is an indicator

**(c)** it is an acid used to relieve acidity **(d)** it is a base used to relieve acidity

**7.** In the same tester the bulb does not glow when a dry wooden pencil case is used. The case is: 1

**(a)** a conductor **(b)** an insulator **(c)** a filament **(d)** a battery

**8.** Which of these metals is most suitable for making a very hot cooking pan? 1

**(a)** copper **(b)** wood **(c)** lead **(d)** sulphur

**9.** A farmer wants to keep tools from rusting during the rainy season. Which is best? 1

**(a)** apply oil and keep dry **(b)** keep in a damp shed

**(c)** wash daily with salt water **(d)** leave outside

**10.** In which year is the average growth rate highest for girls? 1

**(a)** around 40 **(b)** around 11 to 13 years

**(c)** around 20 **(d)** around 2

**11.** Which of the following is a correct application of conduction? 1

**(a)** sunlight **(b)** ceiling fan **(c)** sea breeze **(d)** heating iron rod

**12.** A sprinter's reaction time is 0.2 s and a stopwatch is started after seeing the start. The measured 10 s may be: 1

**(a)** double **(b)** less than true by about 0.2 s
**(c)** more than true **(d)** exact

**13.** Herbivores have a longer small intestine than carnivores because: 1

**(a)** they sleep **(b)** they are bigger
**(c)** they are slower **(d)** plant food takes longer to digest

**14.** A farmer grows wheat after pulses and applies less urea. The reason is: 1

**(a)** urea is costly **(b)** wheat needs no nitrogen
**(c)** pulses consume nitrogen **(d)** nitrogen left by pulses

**15.** A red rose in white light is viewed in a plane mirror. The image of the rose appears: 1

**(a)** red **(b)** green **(c)** white **(d)** black

**16.** A student claims 'the Sun is closer to Earth in summer'. A good test is: 1

**(a)** ask friends
**(b)** look at the Sun
**(c)** compare dates of nearest distance with seasons
**(d)** measure the temperature once

**17.** Liquid A turns China rose pink, liquid B green. If A and B are mixed in equal strengths, the mixture most probably: 1

**(a)** becomes more acidic **(b)** becomes more basic
**(c)** turns almost neutral **(d)** turns blue

**18.** A chain of 10 decorative bulbs goes off when one bulb burns out. The bulbs are connected in: 1

**(a)** open circuit **(b)** parallel **(c)** series **(d)** no circuit

**19.** A metal Z reacts with dilute HCl giving hydrogen but does not react with cold water. Z may be: 1

**(a)** sodium **(b)** zinc **(c)** gold **(d)** potassium

**20.** A solution turns milky when a gas is passed. The gas is probably: 1

**(a)** oxygen **(b)** carbon dioxide **(c)** nitrogen **(d)** hydrogen

**21.** A student with tiredness, pale tongue and poor concentration has these symptoms mostly due to: 1

**(a)** anaemia **(b)** rickets **(c)** scurvy **(d)** goitre

**22.** A candle is placed under a paper spiral hung from a thread. The spiral rotates because: 1

**(a)** heat radiation pushes **(b)** wind from the room
**(c)** hot air rising pushes it **(d)** cold air pushes it

**23.** A man runs round a circular track of length 400 m in 100 s. His average speed is: 1

**(a)** 400 m/s **(b)** 4 m/s **(c)** 40 m/s **(d)** 0 m/s

**24.** A person's stomach acid is removed. Which is the most likely effect? 1

**(a)** faster starch digestion
**(b)** poor protein digestion and more germs reaching the intestine
**(c)** no change
**(d)** better fat digestion

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**25.** In a set-up with a plant in a bell jar with NaOH solution, starch is absent in the leaf because: 1

**(a)** NaOH adds oxygen **(b)** NaOH absorbs carbon dioxide
**(c)** light is blocked **(d)** NaOH gives CO2

**26.** The image formed by a plane mirror of a person 1.6 m tall is: 1

**(a)** 0.8 m tall **(b)** 0.4 m tall **(c)** 3.2 m tall **(d)** 1.6 m tall

**27.** If the Earth revolved around the Sun in half the time, a year would have about: 1

**(a)** 365 days **(b)** 730 days **(c)** 91 days **(d)** 182 days

**28.** A farmer needs 200 kg of lime for 1 hectare of acidic soil. For 2.5 hectares, the lime needed is: 1

**(a)** 250 kg **(b)** 800 kg **(c)** 400 kg **(d)** 500 kg

**29.** In the circuit tester, the bulb glows brightly with a copper wire and dimly with graphite. This shows: 1

**(a)** graphite conducts better **(b)** copper conducts better
**(c)** both conduct equally **(d)** graphite is an insulator

**30.** In the reaction above, the colour of the solution changes from: 1

**(a)** blue to green **(b)** colourless to blue
**(c)** green to blue **(d)** blue to colourless

**31.** A student heats a piece of wax and sees it melt. On stronger heating, black smoke appears. The black smoke indicates: 1

**(a)** melting only **(b)** no change
**(c)** evaporation only **(d)** unburnt carbon, a chemical change

**32.** If testes were removed in an adult, secondary sexual characters may: 1

**(a)** double **(b)** become female
**(c)** stay unchanged forever **(d)** be affected over time

**33.** If the sea and the land were heated equally, the breezes would: 1

**(a)** be reversed **(b)** not form **(c)** be of storms **(d)** be stronger

**34.** A man goes from A to B at 4 km/h and returns at 6 km/h. The time taken in the round trip is 5 h. The distance AB is: 1

**(a)** 12 km **(b)** 6 km **(c)** 5 km **(d)** 24 km

**35.** In one hour this volume is: 1

**(a)** 8 litres **(b)** 480 litres **(c)** 48 litres **(d)** 4800 litres

**36.** Which of these is a net result of photosynthesis in a day? 1

**(a)** less glucose **(b)** no glucose
**(c)** more glucose than used in respiration **(d)** only oxygen

**37.** A student sees a shadow of her hand on a screen 2 m from a lamp. If she moves the hand to the midpoint between lamp and screen, the shadow size relative to the hand is: 1

**(a)** four times **(b)** double **(c)** same **(d)** half

**38.** The Sun is much larger than the Moon, yet in a total solar eclipse the Moon can hide it because the Moon is: 1

**(a)** larger **(b)** much hotter
**(c)** brighter **(d)** much nearer to the Earth

**39.** A tablet that fizzes in water is likely to contain: 1

**(a)** only a base **(b)** a carbonate and an acid
**(c)** only salt **(d)** only sugar

**40.** A tester glows dimly with a pencil lead and brightly with a copper wire. This shows that: 1

**(a)** both conduct, copper better **(b)** copper is an insulator
**(c)** neither conducts **(d)** only graphite conducts

**41.** A student has four metals P, Q, R, S. P and Q react with dilute HCl, R and S do not. Q reacts faster than P. The most reactive is: 1

**(a)** Q **(b)** P **(c)** S **(d)** R

**42.** Why are bicycle chains regularly oiled? 1

**(a)** to make them shiny only **(b)** to make them heavier
**(c)** to make them brittle **(d)** to prevent rusting and reduce friction

**43.** Which of these statements about menarche is correct? 1

**(a)** it never occurs **(b)** it occurs at 60 years
**(c)** it occurs at 5 years **(d)** it can occur from about 10 to 15 years

**44.** Which of the following is a correct application of radiation? 1

**(a)** solar cooker **(b)** melting of ice **(c)** spoon in tea **(d)** heating of milk

**45.** A student measures one oscillation of a pendulum with a stopwatch of reading to 0.1 s and gets 2.0 s. The error is about: 1

**(a)** 50 per cent **(b)** up to 0.5 per cent
**(c)** up to 5 per cent **(d)** nil

**46.** A carnivore's digestive tract compared to a herbivore's is: 1

**(a)** longer **(b)** shorter **(c)** same **(d)** absent

**47.** A student claims that mushrooms make food in sunlight. The correct reply is: 1

**(a)** mushrooms have no chlorophyll and absorb food from dead matter
**(b)** correct
**(c)** they make starch
**(d)** they use CO2

**48.** The colour of the shadow of a red ball in white light is: 1

**(a)** white **(b)** red **(c)** black **(d)** green

**49.** A student says that 'the Moon has phases because the Earth's shadow falls on it'. This is: 1

**(a)** incorrect **(b)** correct
**(c)** partly correct **(d)** correct for the first quarter

**50.** A student adds drops of base to acid and notes the litmus. The sequence of colours is: 1

**(a)** blue, yellow, red **(b)** blue, purple, red
**(c)** red, green, blue **(d)** red, purple, blue

**END OF QUESTION PAPER** **Total: 50 Marks**

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